NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - III

Solution of Initial Value Problems to Ordinary Differential Equation

Syllabus: Numerical solution of ODEs: Solution by Taylor's series - Picard's Method of successive approximations - Euler's and modified Euler's methods - Runge-Kutta methods (second and fourth order).

I. Introduction

Many problems in engineering lead to a **differential equation** whose analytical solution is difficult. Numerical methods give the value of y at chosen points of x.

Initial Value Problem (IVP) : $dy/dx = f(x, y)$ with $y(x_0) = y_0$.
We want y at $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, ... (h = step size).

Two kinds of methods

- **Series / analytical-type** : Taylor series, Picard's method (give y as a function of x).
- **Step-by-step (marching) methods** : Euler, modified Euler, Runge-Kutta (give y at a table of x values).

Sources of error

- **Truncation error** : due to cutting the series / formula after finite terms.

- **Round-off error** : due to limited number of decimal places.
- Smaller h → less truncation error but more steps and more round-off.

2. Taylor's Series Method

Let $y = y(x)$ be the solution of $dy/dx = f(x, y)$, $y(x_0) = y_0$. Expand $y(x)$ about x_0 :

$$y(x) = y_0 + (x-x_0)\,y_0' + (x-x_0)^2/2!\;y_0'' + (x-x_0)^3/3!\;y_0''' + \dots$$

Procedure

- Get $y_0' = f(x_0, y_0)$ from the given equation.
- Differentiate the equation again and again to find y'', y''', $y^{(4)}$ at (x_0, y_0).
- Substitute in the series. Use few terms (up to h^4 is common).
- For the next point, take the new point as starting point (x_1, y_1) and repeat.

Example 1

Solve $dy/dx = x + y$, $y(0) = 1$ and find $y(0.1)$.

$y' = x + y \rightarrow y'(0) = 1$. $y'' = 1 + y' \rightarrow y''(0) = 2$. $y''' = y'' \rightarrow 2$. $y^{(4)} = 2$.

$y = 1 + x(1) + x^2/2\,(2) + x^3/6\,(2) + x^4/24\,(2) = 1 + x + x^2 + x^3/3 + x^4/12$

At $x = 0.1$: $y = 1 + 0.1 + 0.01 + 0.000333 + 0.0000083 =$ **1.11034**

(exact solution $y = 2e^x - x - 1$ gives 1.110342 ✓)

Example 2

Solve $dy/dx = x + y^2$, $y(0) = 1$. Find $y(0.1)$ and $y(0.2)$.

$y' = x + y^2 \to 1$; $y'' = 1 + 2yy' \to 1 + 2 = 3$; $y''' = 2y'^2 + 2yy'' \to 2 + 6 = 8$; $y^{(4)} = 6y'y'' + 2yy''' \to 18 + 16 = 34$

$y = 1 + x + 3x^2/2 + 8x^3/6 + 34x^4/24 + \ldots$

$y(0.1) = 1 + 0.1 + 0.015 + 0.00133 + 0.00014 = \mathbf{1.1165}$

$y(0.2) = 1 + 0.2 + 0.06 + 0.01067 + 0.00227 = \mathbf{1.2729}$

Merits & demerits of Taylor's method

- Merit : gives a continuous solution, accuracy can be improved by adding terms.
- Demerit : needs higher derivatives of $f(x, y)$, which is tedious for complicated functions.

3. Picard's Method of Successive Approximations

Integrate $dy/dx = f(x, y)$ from x_0 to x : $y = y_0 + \int f(x, y)\, dx$ (limits x_0 to x). Start with y_0 and improve:

$$y^{(1)} = y_0 + \int f(x, y_0)\, dx \ , \ y^{(2)} = y_0 + \int f(x, y^{(1)})\, dx$$

$$\boxed{y^{(n)} = y_0 + \int f(x, y^{(n-1)})\, dx}$$

Continue until two successive approximations agree to the required accuracy.

Example 3

Solve $dy/dx = x + y$, $y(0) = 1$ by Picard's method. Find $y(0.1)$.

$f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$.

$y^{(1)} = 1 + \int (x + 1)\, dx = 1 + x + x^2/2$

$y^{(2)} = 1 + \int [\, x + 1 + x + x^2/2 \,]\, dx = 1 + x + x^2 + x^3/6$

$y^{(3)} = 1 + \int [\, x + 1 + x + x^2 + x^3/6 \,]\, dx = 1 + x + x^2 + x^3/3 + x^4/24$

At $x = 0.1$: $y^{(3)} = 1 + 0.1 + 0.01 + 0.000333 + 0.0000042 = 1.11034$

Example 4

Find $y(0.2)$ from $dy/dx = x^2 + y^2$, $y(0) = 0$ (Picard).

$y^{(1)} = \int x^2\, dx = x^3/3$

$y^{(2)} = \int [\, x^2 + x^6/9 \,]\, dx = x^3/3 + x^7/63$

$y(0.2) \approx (0.2)^3/3 + (0.2)^7/63 = 0.002667 + 0.0000002 = 0.002667$

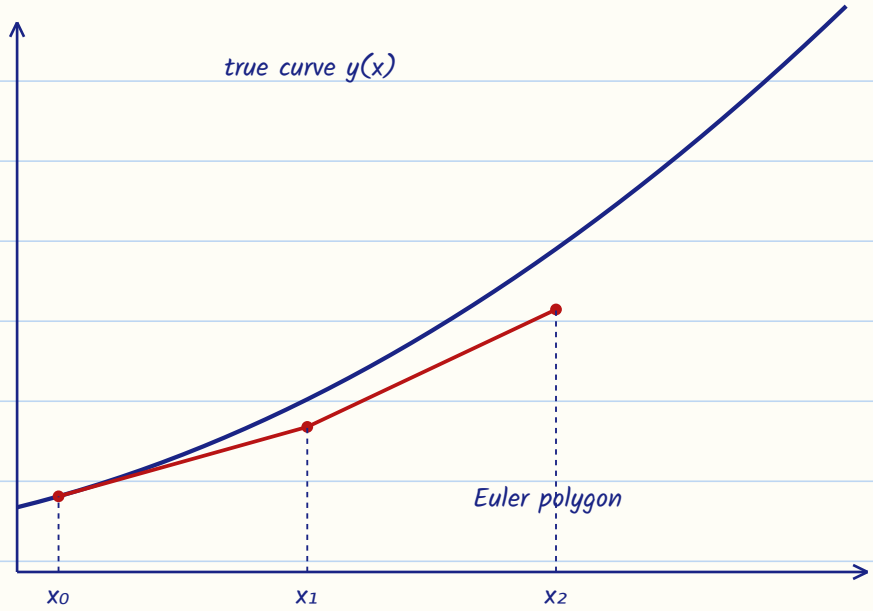
Remarks

- Picard's method gives y as a **function of x**; but integration becomes very difficult for complex $f(x, y)$.
- It is mostly used to prove existence & uniqueness theorem.

4. Euler's Method

Idea : in a small interval h, the curve is replaced by its tangent line at (x_n, y_n).

$$y_{n+1} = y_n + h\, f(x_n, y_n)\ ,\ x_{n+1} = x_n + h$$



$y_1 = y_0 + h\, f(x_0, y_0)$
step h along tangent
error grows at each step
(local error ~ h^2)

It is the simplest, first order method (error per step is of order h^2) but not very accurate.

Example 5

Solve $dy/dx = x + y$, $y(0) = 1$ with $h = 0.1$ for $x = 0.1, 0.2, 0.3$.

n	x_n	y_n	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.1\, f$
0	0.0	1.0	1.0	1.1
1	0.1	1.1	1.2	1.22
2	0.2	1.22	1.42	1.362

So $y(0.1) = 1.1$, $y(0.2) = 1.22$, $y(0.3) = \mathbf{1.362}$. Exact value $y(0.3) = 1.39972$, so error = 0.0377 (large).

Demerits

- Low accuracy, needs very small h; errors accumulate; so we improve it (modified Euler).

5. Modified Euler's Method (Heun's Method)

Average the slopes at both ends of interval. First **predict** with Euler, then **correct** :

$$\text{Predictor} : y_{n+1}^{(0)} = y_n + h\, f(x_n, y_n)$$

$$\text{Corrector} : y_{n+1}^{(1)} = y_n + (h/2)\,[\, f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})\,]$$

The corrector can be repeated till two successive values agree.

Example 6

Same problem $dy/dx = x + y$, $y(0) = 1$, $h = 0.1$.

Step 1 : $y^{(0)}(0.1) = 1 + 0.1(1) = 1.1$. Corrector : $y(0.1) = 1 + 0.05\,[\,1 + (0.1 + 1.1)\,] = 1 + 0.05(2.2) =$ **1.11**

Step 2 : $y^{(0)}(0.2) = 1.11 + 0.1(1.21) = 1.231$. $y(0.2) = 1.11 + 0.05\,[\,1.21 + (0.2 + 1.231)\,] =$ **1.24205**

Step 3 : $y^{(0)}(0.3) = 1.24205 + 0.1(1.44205) = 1.38626$. $y(0.3) = 1.24205 + 0.05\,[\,1.44205 + 1.68626\,] =$ **1.39847**

Much closer to exact 1.39972 than Euler's 1.362.

Example 7

$dy/dx = x + y^2$, $y(0) = 1$, $h = 0.1$. Predictor $y^{(0)}(0.1) = 1 + 0.1(1) = 1.1$. Corrector :

$y(0.1) = 1 + 0.05\,[\,1 + (0.1 + 1.21)\,] = 1 + 0.05(2.31) =$ **1.1155**

(Taylor gave 1.1165)

6. Runge-Kutta Methods

Give high accuracy **without** computing higher derivatives. Agree with Taylor's series up to h^p term (p = order). We need $f(x, y)$ only.

6.1 Second order R-K method

$$k_1 = h\, f(x_n, y_n)\ ,\ k_2 = h\, f(x_n + h, y_n + k_1)$$

$$y_{n+1} = y_n + \tfrac{1}{2}(k_1 + k_2)$$

(identical to modified Euler's method)

Example 8

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$. $k_1 = 0.1(1) = 0.1$; $k_2 = 0.1\, f(0.1, 1.1) = 0.1(1.2) = 0.12$

$y(0.1) = 1 + \tfrac{1}{2}(0.1 + 0.12) =$ **1.11**

6.2 Fourth order R-K method (most used)

$$k_1 = h\, f(x_n, y_n)\ ,\ k_2 = h\, f(x_n + h/2, y_n + k_1/2)$$

$$k_3 = h\, f(x_n + h/2, y_n + k_2/2)\ ,\ k_4 = h\, f(x_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + (1/6)\,[\,k_1 + 2k_2 + 2k_3 + k_4\,]$$

Error of the 4th order R-K method is of order h^5 per step. Four function evaluations per step. Self-starting (needs only y_0).

Example 9

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$. Find $y(0.1)$.

$k_1 = 0.1(0 + 1) = 0.1$

$k_2 = 0.1\, f(0.05, 1.05) = 0.1(1.1) = 0.11$

$k_3 = 0.1\, f(0.05, 1.055) = 0.1(1.105) = 0.1105$

$k_4 = 0.1\, f(0.1, 1.1105) = 0.1(1.2105) = 0.12105$

$y(0.1) = 1 + (1/6)[0.1 + 0.22 + 0.221 + 0.12105] = 1 + 0.662105/6$

$= \mathbf{1.11034}$

Exact = 1.110342 : error less than 10^{-6} !

Example 10

$dy/dx = (y - x)/(y + x)$, $y(0) = 1$, $h = 0.2$. Find $y(0.2)$.

$k_1 = 0.2 \cdot (1 - 0)/(1 + 0) = 0.2$

$k_2 = 0.2\ f(0.1, 1.1) = 0.2\ (1.0/1.2) = 0.1667$

$k_3 = 0.2\ f(0.1, 1.0833) = 0.2\ (0.9833/1.1833) = 0.1662$

$k_4 = 0.2\ f(0.2, 1.1662) = 0.2\ (0.9662/1.3662) = 0.1414$

$y(0.2) = 1 + (1/6)[0.2 + 0.3334 + 0.3324 + 0.1414] = 1 +$

$1.0072/6 = \mathbf{1.1679}$

(Second order R-K for the same problem gives 1.1714.)

6A. More Solved Problems

Example 11 : Euler for a non-linear equation

$dy/dx = x + y^2$, $y(0) = 1$, $h = 0.1$. Find $y(0.1)$, $y(0.2)$, $y(0.3)$.

n	x_n	y_n	$f = x_n + y_n^2$	y_{n+1}
0	0.0	1.0	1.0	1.1
1	0.1	1.1	1.31	1.231
2	0.2	1.231	1.7155	1.4025

So $y(0.3) \approx 1.4025$ by Euler ; the modified Euler method gives 1.4820 for the same data, Taylor's series (4 terms) gives about 1.4825 : Euler is too low here.

Example 12 : Modified Euler, repeating the corrector

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$; find $y(0.1)$ accurate to 4 decimals.

$y^{(0)} = 1.1$. $y^{(1)} = 1 + 0.05\ [\ 1 + 1.2\ ] = 1.1100$

$y^{(2)} = 1 + 0.05\ [\ 1 + (0.1 + 1.11)\ ] = 1.1105$

$y^{(3)} = 1 + 0.05\ [\ 1 + (0.1 + 1.1105)\ ] = 1.11053$

Two successive values agree (1.1105), so **$y(0.1) = 1.1105$** (exact 1.1103).

Example 13 : R-K 4th order, second step

Continue Example 9. Now $x_1 = 0.1$, $y_1 = 1.11034$, $h = 0.1$. Find $y(0.2)$.

$k_1 = 0.1(0.1 + 1.11034) = 0.121034$

$k_2 = 0.1(0.15 + 1.17086) = 0.132086$

$k_3 = 0.1(0.15 + 1.17638) = 0.132638$

$k_4 = 0.1(0.2 + 1.24298) = 0.144298$

$y(0.2) = 1.11034 + (1/6)[0.121034 + 0.264172 + 0.265276 + 0.144298] = 1.11034 + 0.132464 =$ **1.24280**

(Exact value 1.242805 ✓)

Example 14 : Taylor's series in steps

$dy/dx = x - y^2$, $y(0) = 1$. Find $y(0.1)$ by Taylor's series (first 4 terms).

$y' = x - y^2 \to -1$; $y'' = 1 - 2yy' \to 1 + 2 = 3$; $y''' = -2(y'^2 + yy'') \to -2(1 + 3) = -8$

$y = 1 - x + 3x^2/2 - 8x^3/6 + \ldots$

$y(0.1) = 1 - 0.1 + 0.015 - 0.00133 = \mathbf{0.9137}$

Example 15 : Picard for another equation

$dy/dx = 1 + y^2$, $y(0) = 0$. (exact solution $y = \tan x$)

$y^{(1)} = \int (1 + 0)\, dx = x$

$y^{(2)} = \int (1 + x^2)\, dx = x + x^3/3$

$y^{(3)} = \int [1 + (x + x^3/3)^2]\, dx = x + x^3/3 + 2x^5/15 + x^7/63$

Compare with $\tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + \ldots$: first three terms already match.

Derivation of Euler's formula

From Taylor's series $y(x_0 + h) = y_0 + h\, y_0' + (h^2/2)\, y_0'' + \ldots$

Neglect h^2 and higher terms : $y_1 = y_0 + h\, f(x_0, y_0)$. So local truncation error is $O(h^2)$ and global error $O(h)$.

Derivation idea of modified Euler

Integrate $y' = f(x, y)$ from x_n to x_{n+1} and use the **trapezoidal rule** : $y_{n+1} - y_n = (h/2)[\, f(x_n, y_n) + f(x_{n+1}, y_{n+1})\,]$. Since y_{n+1} on the right side is unknown, estimate it by Euler's formula (predictor).

Points to remember while solving R-K problems

- Always compute k_1, k_2, k_3, k_4 **in order**; each uses the previous k.
- Keep 4 to 5 decimal places in the intermediate work.
- To go from x_0 to $x_0 + 2h$, do R-K twice, taking the answer of first step as the new y_0.
- If h is not given, choose $h = (x_final - x_0)/n$, e.g. $n = 1$ or 2.

7. Comparison of the Methods

For $y' = x + y$, $y(0) = 1$, exact $y(0.3) = 1.39972$.

Method	y(0.3)	Error
Euler	1.3620	0.0377
Modified Euler	1.3985	0.0012
R-K 2nd order (h = 0.1)	similar to mod. Euler	small
R-K 4th order	1.39972	$< 10^{-5}$
Method	Order	Remark
Euler	1	simple, poor accuracy
Modified Euler / RK2	2	predictor-corrector
RK4	4	very accurate, popular
Taylor	any	needs derivatives
Picard	any	gives series in x

8. Quick Revision

- Taylor : $y = y_0 + (x-x_0)y_0' + (x-x_0)^2 y_0''/2! + \dots$
- Picard : $y^{(n)} = y_0 + \int f(x, y^{(n-1)})\, dx$
- Euler : $y_{n+1} = y_n + h\, f(x_n, y_n)$
- Modified Euler : $y_{n+1} = y_n + h/2\ [\ f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})\ ]$
- RK4 : $y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)/6$

Exam questions

- Using Taylor's series find $y(0.1)$ for $y' = x^2 + y^2$, $y(0) = 1$.
- Find $y(0.2)$ by Picard's method for $y' = x + y$, $y(0) = 1$.
- Solve $y' = x - y^2$, $y(0) = 1$ for $y(0.2)$ by modified Euler, $h = 0.1$.
- Use R-K method of 4th order to find $y(0.2)$ with $h = 0.1$ for $y' = x + y$, $y(0) = 1$ (answer 1.24281).